

① Unit - 3

- ① Unique aspects of Indian Astronomy
- ② Historical Development of Astronomy in India.
- ③ The Celestial coordinate system.
- ④ Elements of Indian Calendar
- ⑤ Aryabhata & Siddhantic tradition.
- ⑥ Pan ciaraga Astronomical instruments
- ⑦ Jantar Mantar of Raja jai Singh Sawal.

① The Vedanga Jyotisha describes rules for tracking the motions of the Sun & Moon for the purposes of ritual. According to Vedanga Jyotisha in a Yuga era there are 5 solar system years, 167 lunar sidereal cycles, 1,830 day 18335 sidereal days & 62 synodic months.

Astronomy is the oldest of Natural Sciences dating back to antiquity with its origins in the religious, mythological beliefs & practices of Prehistory; vestiges still found in astrology, a discipline long interwoven with Public &

and governmental astronomy. It was not completely separated in Europe during the Copernican Revolution starting in 1543. In some cultures, astronomical data was used for astrological prognostication.

Historical Development of Astronomy in India

- The Vedic living required methods for fixing an auspicious time for performing various rituals & activities. Knowledge of the cardinal directions is also required to set the Vedic altars.
- In ancient times, there were two major wealth-generation activities: Agriculture & Trade. Agriculture was the main driver of the economy & wealth creation. Ability to predict the weather conditions & understand the mechanism of season formation was a key requirement to improve agricultural productivity.
- The other economic activity is related to international trade. This required a good understanding of the latitude, longitude & longitude. Moreover, there was a need for a calendar as well.
- Astronomy plays a crucial role in navigation. The basic idea in navigation is to follow rising or setting location of a particular constellation for a certain period & then turn towards another constellation.

Crucial. Indians have been regularly using astronomy for maritime purposes & some studies suggest that details of this have been well documented.

→ Nilakantha Somayaji in 1500 CE proposed a revised model of planetary motion which closely approximates the mathematical model of planetary motion given by Kepler over a hundred years later in 1609.

Aryabhata developed a mathematical approach to astronomy in his work *Aryabhatiyam*, (499 C.E). This is considered the first full-fledged treatise on mathematical astronomy in India. Varahamihira in his *Pancha-Siddhantika* (~530 C.E) refers to five different astronomical texts which were prevalent during this time. They include Paitamaha, Valistha, Romaka, Paulisa & Saura Siddhantas. They deal with the true motion of the moon, sun diurnal problems, lunar & solar eclipses, movements of planets such as Mercury, Venus, Jupiter & Saturn.

The ⁽⁴⁾ Celestial Coordinates system
" " System is a mechanism for
specifying positions of planet stars & other
celestial objects relative to physical reference
points available to an observer situated
on the earth. It helps to locate a celestial
luminary using a three-dimensional or a spherical
plane.

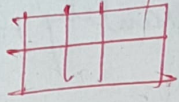
Another useful coordinate system is the Ecliptic is
an imaginary line depicting the sun's movement
around the earth in a path that is tilted with
respect to the spin axis at 23.5°

The horizontal plane with the Zenith & the Nadir
are the two poles of this system. The Zenith is an
imaginary point directly 'above' a particular location
(highest point), on the celestial sphere & nadir
is 180° from Zenith (the lowest point). It helps
to position the stars & other luminaries in the sky
relative to the observer's horizon.

We know that ecliptic is the path
of the sun in the background of stars.
The Moon & this belt is known as 'Zodiac'
or 'Rasicastra'. This Zodiac or Rasicastra
this Zodiac has been divided into 12 equal
parts from a fixed initial point in the ecliptic
in order to trace the trajectory of the Moon
& the other planets in the context of the
stars, known as raśis. Each spans 30° . Shows
the Zodiac signs on the ecliptic.

Magic Square in India

Total will be the same
from every angle



(Add from Your side
Mathematical term)

Unit ③

Unique aspects of Indian Mathematics

→ Mathematics is the science which deals with [the logic of quantity, shape, arrangement.

Maths is all around us.

↓
4000 (thousand) years old concept.

↓
Trigonometry, Algebra, Algorithm, Square root, Cube root, negative numbers, decimal system
9 Indian Mathematicians who transformed the

Forms of knowledge

↓ — Aryabhata - 0

↓ — Brahmagupta — Fibonacci Identity.

sine table

Pythagorean Triples. In 7th

Century, Brahmagupta discovered the first general formula for solving quadratic equations.

↓ — Bhaskara — Indian Mathematician & Astronomer born in Bijapur, Karnataka. He taught how it is applied to astronomical problems & computations is what makes him immensely popular.

not just Calculus, Shastara and
in arithmetic, algebra, the mathematics
of planets & spheres.

4) Hemchandra — Jain Philosopher Scholar,
& an exceptional mathematician. He described Fibonacci.
He also worked with Cadences of
length n. well known as a prodigy
by his generations. He earned
the title Kalikāṭasavajña, i.e.
"the all-knowledge of Kalinyuga".

5) Srinivasa Ramanujan → self learned who coined
the idea of "Taxicab
numbers".

Contribution in analytical
theory of numbers.

1
Elliptic functions

Continued fractions &
infinite series.

6) Satyendra Nath Bose — Developed the Statistical
mechanics for bosons of a
photon gas. He is also
popular for his collaboration
with Albert Einstein. Bose
also worked on Bose-Einstein
Statistics & the theory of Bose-
Einstein condensate.

7) C. R. Rao → American mathematician most popular for his theory of estimation. The Rao Theorem (both related to the quantity of estimators) are his top discoveries. Rao bagged 38 honorary doctoral degrees from universities worldwide in 19 countries & several awards & medals for his contributions to statistics & science.

8) P. C. Mahalanobis — Father of modern statistics in India. Applied statistician. Not only founded the Indian Statistical Institute, he also came up with pioneering studies in anthropometry in India.

9) Shakuntala Devi → Also known as human Computer. Shakuntala Devi competed against a computer in Dallas to give the cube root of 188138577 & she won! She went ahead to compete with the UNIVAC to solve the 23rd root of a 201-digit number & she won that too! She was a woman with outstanding talent & outrageous world records!

~~Anti~~-Arithmetic - Geometry - Trigonometry,
Algebra, Binary Mathematics & Combinational
Problems in Chandah Baskar of Pingala,

Arithmetics → The study of the properties of
traditional operations on numbers - addition,
subtraction, operations on numbers extraction of
roots.

Geometry is with arithmetic, one of the oldest
branches of mathematics who works in the field of
geometry is called a geometer.

Trigonometry → A branch of mathematics that studies
relationships between side lengths &
angles of triangles.

Algebra → The branch of mathematics that helps
in the representation of problems or
situations in the form of mathematical
expressions.

Binary system